

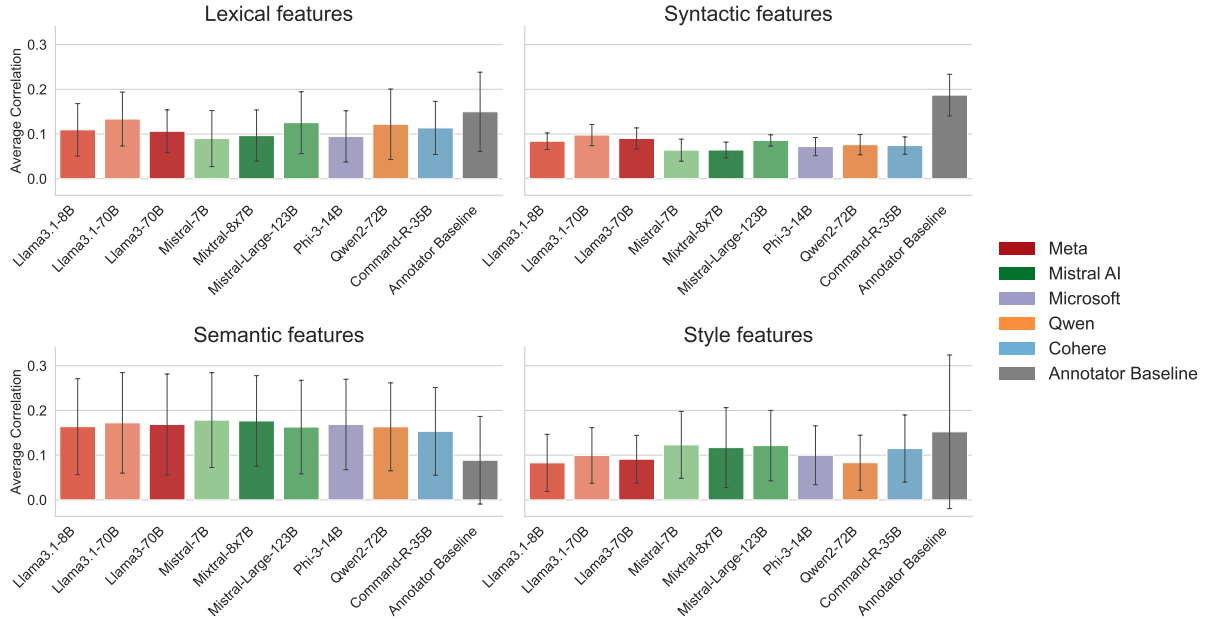


Figure 2: How well do LLMs simulate HUMAN responses to a CHATBOT? We compare the nine models used as SIMULATORS to the original HUMAN by correlating properties of the text they write (Table 1). Bars represent the average correlation across all metrics in a category, and error bars are bootstrapped 95% confidence intervals over these metrics. As a baseline, we also compare the performance of a human annotator on this task. There is limited cross-model variation in performance, and SIMULATORS tend to have higher performance in semantic features and lower performance in syntactic features, while the opposite is true of the human annotators.

distributions include, e.g., the probability that a token with a particular part of speech appears in a sentence across different parts of speech. Finally, feature vectors include semantic (SBERT, Reimers and Gurevych (2019)) and style (LUAR, Soto et al. (2021)) embeddings. In addition, we record if the SIMULATOR and HUMAN each end the conversation on Turn 3 as a binary measure. A summary of all measures included in this study is given in Table 1 in the Appendix.

5 Can LLMs Simulate Human Replies?

In the first experiment, we measure the overall similarity of SIMULATORS to HUMANS (RQ1) by assessing how simulation instructions influence SIMULATOR similarity to HUMAN messages across a variety of models and prompts.

5.1 Experimental Setup

For our first experiment, we use the setup from §4 to generate 828K simulated responses (9 models, 46 prompts, 2,000 English conversations). To evaluate performance, we measure the similarity between the original HUMAN Turn 3 responses and the SIMULATOR Turn 3 responses in lexical, syntactic, semantic, and style domains. Starting with the 21 metrics from §4, we correlate each HUMAN

and SIMULATOR Turn 3 metric. For scalar metrics (e.g., length, classifier probabilities), similarity is the Pearson’s R correlation between HUMAN and SIMULATOR values. For distribution metrics (e.g., POS tags, topics), similarity is the average correlation of HUMAN and SIMULATOR class frequencies over all classes in the distribution. For vector metrics (e.g., BERT, LUAR), similarity is the average correlation of each dimension of the HUMAN and SIMULATOR embedding;¹ since the dimensions of embeddings are not inherently meaningful, we first rotate the embeddings using principal components analysis, so each dimension corresponds to an embedding’s alignment with the direction of a principal component. We then average the correlations for all metrics within each category given in the first column of Table 1 to create four category-wise similarity scores corresponding to lexical, syntactic, semantic, and style. To measure the similarity in ending a conversation, we use the binary F1 score between whether the HUMAN and SIMULATOR decide to end the conversation.

¹A more standard approach for calculating similarity would be to take the average cosine similarity for each HUMAN/SIMULATOR pair. However, we chose to use correlations so the feature vector similarity scores would be comparable to other metrics. Doing so yields similarity scores that are highly correlated with the corresponding cosine similarities.

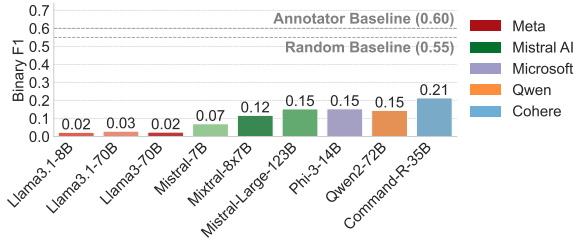


Figure 3: How well do LLMs predict whether HUMANS end a conversation with CHATBOT after the first turn? Each bar represents the binary F1 score of each model predicting whether a conversation will end. The gray horizontal lines show the performance of human annotators and a random baseline that ends the conversation 50% of the time. While there is inter-model variation, all models perform worse than chance. The human annotator performs better than chance.

Although we have a human ground truth for how the conversation did continue, to estimate the task’s difficulty, we also had a separate group of 12 humans who did not author any WildChat data to perform the same task as the SIMULATORS to produce Turn 3 responses. 1273 turns were annotated, and we compared SIMULATOR performance on the subset of turns where both the annotators and SIMULATOR continued the conversation. Full annotation details are in Appendix §C.

5.2 Results

We compare the similarity of different SIMULATORS to the original HUMAN in Figure 2. In general, similarity is lower across all settings for the syntactic measures, while semantic measures tend to be more similar. Additionally, there is little variation across different SIMULATORS. There is effectively no difference between the similarity of SIMULATORS and the similarity of the human annotator baseline across the lexical, semantic, and style categories of measures. However, there is a discernible difference in how similar annotator utterances are syntactically to the original HUMANS compared to the SIMULATORS. Additionally, for both the semantic and style categories, the confidence intervals for the annotator baseline extend beyond 0, the performance we would expect from a totally random baseline, while each SIMULATOR maintains non-random performance. This finding suggests that humans and LLMs have complementary strengths in simulating dialogue; in order to most accurately reflect human utterances, a human-in-the-loop approach where SIMULATORS and annotators play complementary roles may be appro-

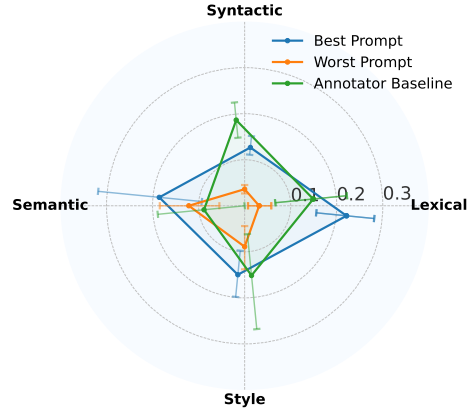


Figure 4: Using the methods from Figure 2, the performance of the best and worst prompts and annotators are compared across metric categories. The best (a DIRECT prompt) and worst (an OVERRIDE prompt) prompts are selected based on an overall average across all metrics and shown in Table 2. The worst prompt underperforms the best prompt in all categories, and annotators outperform all prompts in syntax metrics.

priate. The correlation results for all individual metrics across SIMULATORS are shown in the Appendix in Table 5.

In Figure 3, we compare the similarity between SIMULATORS and HUMANS in their tendency to end the conversation. SIMULATORS seldom end the conversation, continuing 87.1% of the time for COMMAND-R to 98.6% for LLAMA3.1-8B. In contrast, human annotators more accurately mirror the original HUMAN behavior in ending the conversation, indicating that collaboration between SIMULATORS and humans is effective for simulating human interactions. Additionally, human annotators show a better ability to determine when a conversation is likely to end, more often predicting the end at Turn 2 when it actually occurs, whereas SIMULATORS tend to predict conversation end with similar frequency regardless of the HUMANS actual behavior (Figure 11). The performance differences across SIMULATOR models are mainly driven by how frequently each model ends the conversation.

Finally, we look at the impact of prompt on SIMULATOR and HUMAN similarity in Figure 4. We plot the average similarity within each category for the best prompt (i.e., the highest average correlation across all metrics), worst prompt, and human annotator baseline. See prompts in Table 2. The best prompt shows higher lexical and semantic similarity than the human annotator baseline, while human annotators have higher syntactic similarity, in line with the results on models in Figure 2. We

also see that prompts significantly affect the similarity across all categories, suggesting that prompt engineering impacts more than model selection for simulating human interactions. This effect is robust based on the error bars of each method. The most impacted categories are the lexical and syntactic categories, while the prompt has less effect on semantic and stylistic similarity. Given that models also demonstrate little difference in these categories, it is challenging to engineer human simulations using LLMs that are semantically or stylistically similar, while it is possible with good prompt engineering to improve lexical and syntactic similarity. The correlation results of all metrics across prompts are shown in Appendix Table 6.

Overall, we find that the choice of SIMULATOR has only a minor impact on the ability to simulate HUMANS, while the design of the prompt is most relevant when optimizing performance (RQ1).

6 Simulation in non-English Languages

Next, we address how the results from §5 generalize to languages beyond English (RQ2).

6.1 Experimental Setup

We compare the performance of SIMULATORS across three languages: English, Chinese, and Russian. Chinese and Russian were selected for their substantial representation in the WildChat dataset, following English. Specifically, Chinese and Russian comprise 15.9% and 10.5%, respectively, whereas English accounts for 53%. For our analysis, we randomly sample 10,000 conversations from each language.

To generate conversations, we use a subset of models from the previous section: MISTRAL-LARGE-123B, LLAMA3.1-70B, and MIXTRAL-8X7B. We identify the most effective English prompts from each prompt category and have these prompts manually translated into Chinese and Russian by native speakers. We select three prompts that consistently perform well across six largely uncorrelated metrics: capitalization, punctuation, part of speech, SBERT embeddings, sentiment, and politeness. (Appendix §E, Figure 9). The selected prompts are presented in Appendix Table 3.

To measure the similarity between HUMAN and SIMULATOR responses, we use a procedure similar to the one described in §5. Due to the unavailability of several metrics from §5 for Chinese and Russian, we focus on a subset of ten metrics that

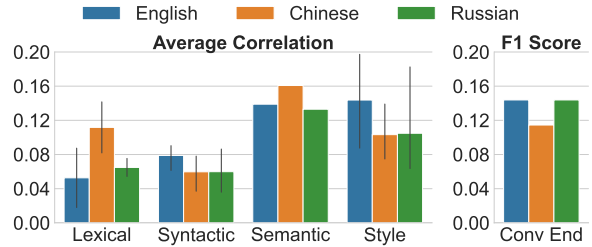


Figure 5: How well do SIMULATORS replicate HUMAN text across languages? Similar to Figures 2 and 3, we plot the similarity between SIMULATOR and HUMAN text across ten metrics in three languages. English, Chinese, and Russian have similar performance patterns across all five categories of metrics. However, some differences exist (e.g., Chinese SIMULATORS outperform other languages in lexical and semantic metrics but underperform in conversation endings). Correlations of individual metrics are shown in Tables 7 and 8.

cover all four categories. Detailed descriptions of these metrics and their operationalizations can be found in Appendix Table 4. We employ consistent or similarly trained models whenever possible to ensure comparability across languages.

6.2 Results

The aggregated metrics for the three languages are depicted in Figure 5. Consistent with the findings discussed in §5, all three languages show higher correlations between SIMULATORS and HUMANS in the semantic metric but lower correlations in the syntactic metric. For detailed correlations of individual metrics, we refer to Appendix Table 7 and Table 8. Additionally, predicting whether a conversation will end consistently performs below chance across all languages, as SIMULATORS are unlikely to predict a conversation will end. These observations reinforce the conclusions drawn in §5.

Notably, there are differences between the languages. Chinese SIMULATORS, for example, outperform their English and Russian counterparts in lexical metrics and show slightly better performance in semantic metrics. The differences in lexical metrics, such as utterance length and perplexity, may be attributed to the typically shorter sentence lengths in Chinese. Conversely, English SIMULATORS excel in predicting style metrics compared to those in other languages. Toxicity and sentiment metrics primarily contribute to this improvement. In English, SIMULATORS more accurately reflect the variations in toxicity and sentiment of HUMANS. This capability may vary across languages because Chinese and Russian have rel-